WORKED SOLUTION

STEP III 2017, Question 12

Probability and Statistics • Joint Discrete Distributions

STEP III 2017, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s3-q12

The key idea

Everything follows from two standard sums, $\sum_{r=1}^{n} r = \frac{1}{2}n(n+1)$ and $\sum_{r=1}^{n} r^2 = \frac{1}{6}n(n+1)(2n+1)$. First fix k by making the probabilities add up to 1. Then find the marginal distribution by summing over y. For the covariance we need $\mathrm{E}(X)$ and $\mathrm{E}(XY)$. The symmetry of $x+y$ in x and y halves the work.

Solution

Part (i)

The probabilities over all n^2 pairs (x,y) must add up to 1. By symmetry, $\sum_{x,y} x = \sum_{x,y} y$. Each of these equals $n\sum_{x=1}^{n} x = \frac{1}{2}n^2(n+1)$, since every x appears once for each of the n values of y. So

$$1 = k\sum_{x=1}^{n}\sum_{y=1}^{n}(x+y) = k\cdot 2\cdot \tfrac{1}{2}n^2(n+1) = k\,n^2(n+1), \qquad \text{so} \qquad k = \frac{1}{n^2(n+1)}.$$

For the marginal distribution of X we sum over y, with x fixed:

$$\begin{aligned}\mathrm{P}(X=x) &= k\sum_{y=1}^{n}(x+y) = k\Big(nx + \tfrac{1}{2}n(n+1)\Big) = \frac{kn\,(2x+n+1)}{2}\\ &= \frac{n+1+2x}{2n(n+1)},\end{aligned}$$

which is the required result. Since $x+y$ is symmetric in x and y, Y has the same distribution: $\mathrm{P}(Y=y) = \dfrac{n+1+2y}{2n(n+1)}$.

Ask yourself To show that X and Y are *not* independent, do we have to check every pair (x,y)?

No. Independence needs $\mathrm{P}(X=x, Y=y) = \mathrm{P}(X=x)\,\mathrm{P}(Y=y)$ for *every* pair, so one pair where it fails is enough. Take $x=y=1$:

$$\mathrm{P}(X=1,Y=1) = 2k = \frac{2}{n^2(n+1)}, \qquad \mathrm{P}(X=1)\,\mathrm{P}(Y=1) = \frac{(n+3)^2}{4n^2(n+1)^2}.$$

These are equal exactly when $8(n+1) = (n+3)^2$, that is $n^2 - 2n + 1 = 0$, or $(n-1)^2 = 0$. So they are equal only for $n=1$. Since $n \geq 2$, the two probabilities differ, and X and Y are **not independent**.

Part (ii)

We use $\mathrm{Cov}(X,Y) = \mathrm{E}(XY) - \mathrm{E}(X)\,\mathrm{E}(Y)$, and note that $\mathrm{E}(Y) = \mathrm{E}(X)$ because the two variables have the same distribution.

The mean. Using the marginal from part (i),

$$\begin{aligned}\mathrm{E}(X) &= \frac{1}{2n(n+1)} \sum_{x=1}^{n} ((n+1)x + 2x^2) = \frac{1}{2n(n+1)} \Big(\tfrac{1}{2} n(n+1)^2 + \tfrac{1}{3} n(n+1)(2n+1) \Big) \\ &= \frac{1}{2} \Big(\frac{n+1}{2} + \frac{2n+1}{3} \Big) = \frac{7n+5}{12}.\end{aligned}$$

The mean of the product. Here $xy(x+y) = x^2y + xy^2$, and the two parts give equal double sums by symmetry. Also, the double sum of $x^2 y$ factorises as a product of single sums:

$$\begin{aligned}\mathrm{E}(XY) &= k \sum_{x=1}^{n} \sum_{y=1}^{n} xy(x+y) = 2k \Big(\sum_{x=1}^{n} x^2 \Big) \Big(\sum_{y=1}^{n} y \Big) = 2k \cdot \frac{n(n+1)(2n+1)}{6} \cdot \frac{n(n+1)}{2} \\ &= \frac{n^2(n+1)^2(2n+1)}{6\,n^2(n+1)} = \frac{(n+1)(2n+1)}{6}.\end{aligned}$$

The covariance. Putting these together over the common denominator 144:

$$\begin{aligned}\mathrm{Cov}(X,Y) &= \frac{(n+1)(2n+1)}{6} - \frac{(7n+5)^2}{144} = \frac{24(2n^2+3n+1) - (49n^2+70n+25)}{144} \\ &= \frac{-n^2+2n-1}{144} = -\frac{(n-1)^2}{144}.\end{aligned}$$

Since $n \geq 2$, $(n-1)^2 > 0$, so $\mathrm{Cov}(X,Y) = -\dfrac{(n-1)^2}{144} < 0$. The covariance is negative, as required.

It makes sense that the covariance is negative. The joint probability depends only on $x+y$, which encourages a large total. If X is already large, Y does not need to be large as well to make the total large, so Y has slightly less reason to be big.

Checks

- Take $n=2$. Then $k = \frac{1}{12}$, and the four probabilities are $\frac{2}{12}, \frac{3}{12}, \frac{3}{12}, \frac{4}{12}$, which add up to 1. The marginals are $\mathrm{P}(X=1) = \frac{5}{12}$ and $\mathrm{P}(X=2) = \frac{7}{12}$, matching the formula.

So $\mathrm{E}(X) = \frac{19}{12}$ and $\mathrm{E}(XY) = \frac{2+6+6+16}{12} = \frac{5}{2}$. This gives $\mathrm{Cov}(X,Y) = \frac{5}{2} - \frac{361}{144} = -\frac{1}{144}$, which agrees with $-(n-1)^2/144$.

- The marginal probabilities add up to 1: $\sum_x (n+1+2x) = n(n+1) + n(n+1) = 2n(n+1)$.
- The value of $(n-1)^2$ that makes the independence test fail is the same one that makes the covariance zero. In both, $n = 1$ is the degenerate case where $X = Y = 1$ with certainty.
- A computer check of the exact distributions for $n = 2, \ldots, 29$ confirmed k, the marginals, $\mathrm{E}(X)$, $\mathrm{E}(XY)$ and the covariance formula.

Common slips

- Forgetting that $\sum_{x,y} x$ has a factor n from the sum over y. This gives $k = 1/(n(n+1))$, which is wrong.
- Trying to show non-independence "in general" rather than with one explicit pair. A single counterexample is both enough and the clearest argument.
- Thinking that a nonzero covariance is needed to show dependence. Here it does show dependence, but zero covariance would *not* prove independence.

Beyond STEP

A joint distribution of the form $f(x,y) = g(x+y)$ is a simple model of variables that are linked through their total. The sign of the covariance then depends on the shape of g. Here g is linear, and the covariance is a small negative quantity. At university, joint distributions, marginals, conditional distributions and correlation are studied together, with continuous versions using double integrals in place of double sums. Grimmett and Welsh, *Probability: An Introduction*, covers this material in a style close to STEP.

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